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**Department : B.TECH(AIDS)**

**Subject : Problem Solving Using Python**

**Batch : 03**

**Project Title: Quiz Game with Score Analyzer(With Fundamentals)**

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**AIM**

To develop a Python program that conducts a quiz, evaluates user responses, calculates score and percentage, and demonstrates fundamental programming concepts such as loops, decision making, functions, and recursion.

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**ALGORITHM**

1. Start the program
2. Store questions and answers in a list of tuples
3. Initialize score = 0
4. Display each question using a loop
5. Get user input for each question
6. Compare user answer with correct answer
    - If correct → increase score
    - Else → display correct answer
7. After all questions:
    - Calculate percentage
    - Display performance (Excellent / Good / Needs Improvement)
8. Display unique answers using set
9. Perform string operations (upper, split)
10. Ask user to retry:

- If yes → restart using recursion

- Else → exit

11. Stop program

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**CODE**

```
questions = [
    ("What is 2 + 2?", "4"),
    ("Capital of India?", "Delhi"),
    ("Python is a (language/program)?", "language"),
    ("5 * 6 = ?", "30")
]

def play_quiz():
    score = 0

    for q, ans in questions:
        user = input(q + " ").strip()

        if user.lower() == ans.lower():
            print("Correct!\n")
            score += 1
        else:
            print(f"Wrong! Answer is {ans}\n")

    return score


def analyze(score):
    total = len(questions)
    percent = (score / total) * 100

    print(f"\nScore: {score}/{total}")
    print(f"Percentage: {percent:.2f}%")

    if percent >= 75:
        print("Excellent")
    elif percent >= 50:
        print("Good")
    else:
```

```
        print("Needs Improvement")


def remove_duplicates():
    answers = [ans for _, ans in questions]
    unique = set(answers)

    print("Unique Answers:", unique)


def string_demo():
    text = input("Enter any sentence: ")
    print("Upper:", text.upper())
    print("Split:", text.split())


def retry():
    choice = input("\nPlay again? (yes/no): ").lower()

    if choice == "yes":
        main()
    elif choice == "no":
        print("Thanks for playing!")
    else:
        print("Invalid input")
        retry()


def main():
    print("\n--- Quiz Game ---")

    score = play_quiz()
    analyze(score)
    remove_duplicates()
    string_demo()
    retry()


main()
```

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**SAMPLE OUTPUT**

```
C:\Users\Nikash> python quiz_project.py

--- Quiz Game ---

What is 2 + 2? 4
Correct!

Capital of India? Delhi
Correct!

Python is a (language/program)? language
Correct!

5 * 6 = ? 25
Wrong! Answer is 30


Score: 3/4
Percentage: 75.00%
Excellent

Unique Answers: {'Delhi', '4', 'language', '30'}

Enter any sentence: python is fun
Upper: PYTHON IS FUN
Split: ['python', 'is', 'fun']

Play again? (yes/no): no
Thanks for playing!
```

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**RESULT**

The program was successfully executed. The quiz system correctly accepted user inputs, evaluated answers, calculated scores and percentages, and demonstrated fundamental programming concepts such as loops, conditions, functions, sets, strings, and recursion.